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Q1. Harshad number

ANSWER

```
def is_harshad(n):  
    s = 0  
    temp = n  
    while temp > 0:  
        s += temp % 10  
        temp //= 10  
    return n % s == 0  
  
n = int(input("Enter a number: "))  
  
if is_harshad(n):  
    print("Harshad Number")  
else:  
    print("Not a Harshad Number")  
  
print("Harshad numbers between 1 and 500:")  
for i in range(1, 501):  
    if is_harshad(i):  
        print(i, end=" ")
```

Q2. GCD

CODE

```
a = int(input("Enter first: "))  
b = int(input("Enter second: "))  
  
while b != 0:  
    temp = b  
    b = a % b  
    a = temp  
  
print("GCD is", a)
```

INPUT

```
12  
18
```

OUTPUT

```
Enter first: Enter second: GCD is 6
```

Q3. sum to a target value

ANSWER

1. 'set'
2. 'lst'
3. 'num'
4. 'seen'
5. 'seen'
6. 'pairs'
7. 'find_pairs'

Q4. Bitwise Operators

ANSWER

```

a = 13 (int)
b = 21 (int)
c = 26 (int)

result = (a & b) | c
        = (13 & 21) | 26
        = 5 | 26
        = 31 (int)

```

Output:

```

a=13, b=21, c=26, result=31
0b11111 0o37 0x1f
7 62 -32

```

Q5. binary search

CODE

```

def rotated_binary_search(arr, target):
    low = 0
    high = len(arr) - 1

    while low ≤ high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid

        # Check if left half is sorted
        if arr[low] ≤ arr[mid]:
            # Target is in left sorted half
            if target ≥ arr[low] and target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1
        # Else right half is sorted
        else:
            # Target is in right sorted half
            if target > arr[mid] and target ≤ arr[high]:
                low = mid + 1
            else:
                high = mid - 1

    return -1 # not found

data = [4, 5, 6, 7, 0, 1, 2]
t = int(input("Target: "))
result = rotated_binary_search(data, t)

if result ≠ -1:
    print("Found at:", result)
else:
    print("Not Found")

```

INPUT 6

OUTPUT Target: Found at: 2